higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FUNDAMENTALS
NQF LEVEL 2

8 MARCH 2019

This marking guideline consists of 5 pages.

Copyright reserved

QUESTION 1

1.1 Engineering and related design
Examples:
- Agricultural engineering
- Chemical engineering
- Civil engineering
- Electrical engineering
- Mechanical engineering (Any TWO examples)

Manufacturing and Assembly
Examples:
- Auto Assembly
- Baking
- Graphic arts and printing
- Metal products
- Textile products (Any TWO examples)

Fabrication and Extraction
Examples:
- Construction material
- Ore extraction
- Primary plastics
- Textiles
- Primary rubber and latex (Any TWO examples)

(3 × 3) (9)

1.2 Career is a job or profession you have been trained for (1)

1.3
1.3.1 D
1.3.2 E
1.3.3 A
1.3.4 B
1.3.5 L
1.3.6 J
1.3.7 F
1.3.8 I
1.3.9 K
1.3.10 H

(10 × 2) (20)

1.4 1.4.1
- To support the rights and responsibilities of employees and of the employers with regard to fair labour practices
- To make sure that disciplinary action is carried out in a fair and correct way
- To promote peace in the workplace, by applying the Constitution and the Bill of Rights
- To encourage employees to take part in the decision-making process
- To allow workers to join trade unions
- To allow employers to belong to an employer's organisation
- To put bargaining councils in place to negotiate conditions such as wages between employers and employees. (7)

1.4.2 Basic Conditions of Employment Act gives employees a right to fair labour practices.✓ It does this by laying down rules (the most important sections) for the Basic Condition of Employment Act✓ such as hours of work, overtime pay, sick leave and annual leave.✓ (3)

[40]

QUESTION 2

2.1
- Safety signs are intended to identify hazards.
- To provide emergency information.
- To direct people to emergency exits in the workshop. (3)

2.1 2.2.1 Warning signs
2.2.2 Yellow and black
2.2.3 White and red
2.2.4 A sign prohibiting certain behaviour
2.2.5 Safe condition sign
2.2.6 A sign providing information about safe condition
2.2.7 Blue
2.2.8 A sign indicating that a special course of action is required
2.2.9 Supplementary
2.2.10 A sign with text, it can be used in conjunction with a safety sign to provide additional information

(10 × 1) (10)

2.3
- Personal safeguarding
- Machinery safeguarding
- Electrical safeguarding (3)

2.4
- Do not play the fool when operating any machine.
- Use all guards and safety equipment provided.
- Never use machine without permission or correct training.
- Never touch rotating blades or cutters.
- There must be one operator at a time.
- Never lean or sit on a machine
- Never clean or adjust a machine without switching it off first. (Any 4 × 1) (4)

2.5
- Overalls and closed shoes must be worn
- Respiratory protection
- Latex gloves/long rubber gloves
- Safety goggles
- Face mask
- Apron (Any 5 × 1) (5)

2.6
- Know how to deal with emergencies.
- Always wear proper PPE.
- Read all safety information and labels.
- Understand how the chemicals you work with can affect your health.
- Do not allow any sparks, flames or hot surfaces anywhere near dangerous chemicals.
- Keep only the smallest amount of chemicals.
- Follow the health and safety rules of your workplace. (Any 5 × 1) (5)

[30]

QUESTION 3

3.1
- Electrical faults
- Open flames
- Hot surfaces
- Chemical reactions
- Friction
- Static electricity (Any 5 × 1) (5)

3.2
- Clean up any spills of flammable liquids as soon as you see them.
- Label and store flammable liquids carefully.
- Never light a match or cigarette or have any open flame near flammable liquids. (3)

3.3 The procedure is to sound (set) the alarm✓ and send for assistance before attempting to fight a fire.✓ (2)

[10]

QUESTION 4

4.1
- If the injury seems serious, send for medical aid immediately.
- Do not move the person unless absolutely necessary. You may cause further injury.
- Remain calm and offer reassurance and comfort.
- Check the person for emergency medical information tags or cards. (4)

4.2
- History
- Symptoms
- Signs (3)

[7]

QUESTION 5

5.1
5.1.1 Monthly
5.1.2 Monthly
5.1.3 12 Months
5.1.4 Yearly
5.1.5 Yearly

(5 × 1) (5)

5.2
- Appoint committee members
- Attend every committee meetings
- Ensure that at least they meet at least every three months
- Decide on the number of health and safety committee members (4)

5.3
5.3.1 A person who is employed, work for an employer and who receives a numeration.

5.3.2 Any premises or place where a person performs work in the course of his employment.

(2 × 2) (4)

[13]

TOTAL: 100

Copyright reserved